LSTM Neural Networks Anomaly Detection for Biotic and Abiotic Early Stress Detection on Tomato Plants

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***Abstract*—This study explores time series anomaly detection using a long short-term memory (LSTM) neural network to identify abnormal plant conditions—both biotic and abiotic—by analyzing a stem frequency parameter. This parameter, closely linked to plant stem conductivity, serves as a novel indicator of plant hydration and overall physiological status. Four independent experiments were conducted between 28 May 2024, and 16 December 2024, and 20 tomato plants were maintained under different conditions (healthy, biotic, and abiotic stress) to assess physiological differences and develop models for early symptom detection. We used samples from healthy plants to train an LSTM neural network to predict stem frequency values 24 h ahead. The trained model was then used to forecast the behavior of stressed plants, and an anomaly was flagged whenever the predicted values significantly deviated from the measured ones. Our LSTM-based anomaly detection model successfully detected water stress conditions several days before visible symptoms appeared. However, the algorithm struggled to distinguish early signs of Fusarium infection. Despite this limitation, in most cases, the model provided early warnings of biotic stress before any visual symptoms were evident. Future research will focus on expanding the dataset to enhance the model's ability to differentiate between various types of plant stress. This article builds upon the work presented at CAFE 2024 Cum et al., 2024, utilizing the same experimental setup but exploring a completely different approach for the early identification of stress symptoms in tomato plants.**

***Index Terms*—Drought, food security, Fusarium wilt, plant abiotic stress, smart agriculture.**

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I. Research Context and Introduction

Agriculture is deeply influenced by various environmental factors that affect crop growth and productivity. Plant stresses can be divided into two main categories: biotic and abiotic stress. Biotic stresses are caused by living organisms, such as pests, diseases, and weeds, which can significantly reduce crop yields and quality [2], [3]. On the other hand, abiotic stresses arise from nonliving environmental factors, such as drought, salinity, extreme temperatures, and nutrient deficiencies [4]. The interplay of these stressors, combined with challenges related to global warming and increased food demand due to overpopulation, poses a significant risk to food security. In this scenario, smart agriculture is a promising solution to overcome such problems. Plant yield is deeply influenced by external conditions that may cause plant stress (such as heat extremes and water scarcity) or biotic causes, such as pests and pathogens. In this context, efficient resource management and proactive monitoring of plants are crucial to guarantee better food production and more efficient resource management. Recent advancements in machine learning have introduced powerful techniques that can significantly enhance the agricultural sector's productivity and efficiency. For example, Barradas et al. [5] used reflectance spectroscopy on leaves and different machine learning models to automatically detect drought conditions in *Arabidopsis thaliana*. Deep learning techniques are widely used in literature to detect pests and diseases [6]. Picon et al. [7] used state-of-the-art convolutional neural networks to detect visual symptoms of diseases on different crop types. In the context of plant stress detection, stem electrical impedance is emerging as a novel and promising parameter. Various researchers have highlighted its potential in recent studies. For instance, Garlando et al. [8] demonstrated significant differences in stem impedance measurements when tobacco plants were exposed to varying watering conditions. Furthermore, studies such as those by the authors in [9] and [10] have explored stem electrical impedance as a feature for binary classifiers capable of detecting general stress in plants. Another interesting application for stem impedance measurements is detecting iron deficiency in tomato plants, as highlighted by Hamed et al. [11], which provided promising results.

This study aims to identify biotic and abiotic stress symptoms in tomato plants using stem frequency, a parameter able to provide insights into plant hydration and overall health before the visible symptoms appear on the plant (such as wilting or yellow leaves). The research focuses on two types of plant stress: abiotic stress, induced by water deprivation, and biotic stress, caused by infection with the pathogenic fungus *Fusarium oxysporum*. Tomato is one of the most widely grown plants producing nongrain products with a worldwide production exceeding 180 million tonnes [12] worldwide, and pathogens such as *Fusarium* pose significant challenges in tomato production worldwide, leading to a substantial decrease in production, for example, it is estimated that in India Tomato production reduced by 45% due to Fusarium wilt [13], the primary disease caused by this pathogen. Fusarium oxysporum is primarily transmitted through infected soil, plant debris, contaminated seeds, irrigation water, root-to-root contact, and farm equipment, allowing spread in agricultural environments. This disease leads to different visible symptoms, such as progressive leaf yellowing, wilting, and necrosis of vascular tissues. Moreover, the infection blocks xylem vessels, preventing water transport and causing plant wilting, leading to the death of the plant. Biotic stresses such as Fusarium wilt, combined with the increasingly frequent heat waves and droughts caused by global warming, make early identification of the problem imperative to prevent significant production losses.

This study employs an anomaly detection approach [14] to detect stress-related anomalies. Anomaly detection includes various processes and algorithms for identifying patterns or data that significantly deviate from expected behavior. It is widely used in fields such as network security, fraud detection, and predictive maintenance, with statistical methods and machine learning being extensively employed to enhance the efficiency and accuracy of these anomaly detection systems, but many applications can be found in the growing field of smart agriculture [15]. In this context, long short-term memory (LSTM) neural networks are widely used to predict the expected behavior of a series of data [16]. The key idea in this article is to predict the expected behavior of a plant parameter, the stem frequency. If it deviates, the algorithm can detect abnormal behavior.

Plants maintained under optimal conditions are used to train an LSTM neural network, which models the expected behavior of stem frequency in healthy plants. The trained model predicts stem frequency values for the following day at the same hour. These predicted values are then compared to actual measurements: an anomaly is flagged if the prediction error exceeds a predefined threshold or the exact value falls below a domain knowledge-based threshold.

This article is an extension of the work conducted in [1]. In particular, a larger dataset is constructed, and a completely different analysis is performed.

This research article is organized as follows.

1) Section II details the experimental setup for data collection and plant management.
2) Section III describes the methodology for data preprocessing and the algorithms employed.
3) Section IV presents and analyzes the results.
4) Section V concludes the article and outlines the key findings and implications.

II. Experimental Setup

In this study, **20 tomato plants (Solanum lycopersicum)**, initially three weeks old, were transplanted into 3.6 liter pots. The plants were kept in a greenhouse under natural light, where the average temperature during the day was 25 °C and the temperature at night was 18 °C. Relative air humidity averaged 56% during the day and 61% at night.

Different experiments were conducted, in particular, as follows.

1) *Experiment 1:* From 28 May 2024 to 4 July 2024.
2) *Experiment 2:* From 22 July 2024 to 14 September 2024.
3) *Experiment 3:* From 21 September 2024 to 21 October 2024.
4) *Experiment 4:* From 24 October 2024 to 16 December 2024.

Irrigation was carried out by filling the saucer pots, and artificial infection was induced by adding **5 g/l** of Fusarium sample to the soil after the first week of measurements.

In total, **20 plants** were analyzed under different conditions as follows.

1) *Eight plants (Healthy condition):* Regular watering, no fungal infection.
2) *Eight plants (Drought condition):* Complete water deprivation, no fungal infection.
3) *Four plants (Fusarium infection):* Regular watering, artificially infected with the Fusarium fungus.

The plants were monitored during all the experiment period annotating the start of visible symptoms due to the applied stressors.

A. Sensors Measurements

The plants were monitored using the system developed in [17]. The sensors are based on a relaxation oscillator, where the plant stem is part of the circuit. Variations in stem conductivity cause shifts in oscillator frequency, which is continuously measured by an **STM32WL55JC1 microcontroller**. The recorded data are transmitted wirelessly via the **LoRa protocol** over the free ISM band (868 MHz in Europe). This approach can be seen as an indirect measurement of the stem electrical conductivity, which can give useful insights about plant hydration conditions and its overall health. In a first approximation, a more hydrated plant offers a higher conductivity hence a bigger stem frequency, and vice versa. Two surgical needles were used as electrodes to ensure proper electrical contact, positioned **2-cm apart**. Their small size was necessary due to the reduced diameter of the plant's stems, and their biocompatibility ensured minimal physiological impact on the subjects.

An illustration of electrode placement is shown in Fig. 1.

In addition to the stem frequency measurement system, soil water potential [18] was measured to assess the hydration status of the plant's surroundings [19] helping the learning algorithms to discriminate biotic and abiotic stress. For this purpose, the watermark sensor [20] was selected. This sensor operates on a resistive principle, where changes in soil water content alter its electrical resistance. The same method used for stem

Fig. 1. Sensor connection with 3-D-printed support for precise needles positioning.

frequency measurement was applied to read the watermark sensor: a relaxation oscillator, implemented using an LMC555 timer, generates a square wave whose frequency depends on the sensor's resistance. As soil moisture levels vary, the watermark sensor's resistance changes accordingly, leading to a measurable shift in oscillation frequency.

Overall, throughout the acquisition period, a total of 42 053 samples were collected across all experiments.

III. Methodology

This section provides a detailed description of the methods used in this research. In particular, the data preprocessing step and the statistical tests conducted to evaluate the relevance of the selected features and the algorithms employed for anomaly detection are outlined.

A. Data Preprocessing

The data acquired across all experiments underwent a preprocessing step, as is standard in a typical machine learning pipeline. The first crucial aspect to consider is the intrinsic variability among different plants. During previous studies, individual plants exhibited different baseline stem frequency values, likely attributable to inter-plant variability in stem conductivity [1]. Such variability may be considered a between-subject difference in the same way for human patients

To account for this, the initial preprocessing step involved establishing a baseline value for each plant. The first week of data acquisition was designated as a calibration period under the assumption that all plants were healthy during this time. Stem frequency measurements collected during this period were averaged to determine a baseline value for each plant. The baseline stem frequency was computed for each tomato plant by averaging measurements taken at a sampling rate of one acquisition every 60 min. This resulted in a total of 168 samples per plant during first week of acquisition. Table I presents the baseline values for different plants.

After acquiring the baseline, two features were extracted from the stem frequency parameter as follows.

- *Relative frequency change (RFC):* The relative variation in stem frequency concerning the initial baseline, calculated as follows:

$$\mathrm{RFC}_k = \frac{\mathrm{StemFreq}_{[k]} - \mathrm{Baseline}}{\mathrm{Baseline}}. \tag{1}$$

- *Stem frequency slope (first order difference):* The rate of change of the relative frequency variation during 1 h, since one sample is recorded every hour the result is the difference between two consecutive stem frequency samples

$$\mathrm{Slope} = \frac{\mathrm{StemFreq}_{[k]} - \mathrm{StemFreq}_{[k-1]}}{\mathrm{One\ sample}}. \tag{2}$$

In addition, a normalization step is applied to the soil water potential measurement. Since the employed sensor is reliable in a range from 0 to –200 kPa, the SWP measurement was normalized as follows:

$$\mathrm{SWP}_{\mathrm{norm}} = \frac{\mathrm{SWP}}{200\ \mathrm{kPa}}. \tag{3}$$

B. LSTM and Anomaly Detection

LSTM neural networks are recurrent neural networks (RNNs) designed to handle sequential data, making them suitable for time series prediction. The LSTM addresses the problem of the vanishing gradients affecting traditional RNNs. The LSTM neural network comprises a cell and three gates: the input, output, and forgets gates. The cell can store information over time intervals, and the gates manage the data flow in and out. The forget gate is responsible for deciding which past information is irrelevant; hence, it is discarded. Similarly, the input gate decides which new information the network has to use. This selective process allows LSTMs to maintain long-term memory and predict across multiple time steps ahead. In this work, we decided to use LSTM networks to learn the temporal pattern of stem frequency for a healthy plant. Then, the network is applied to unseen stressed plants, and anomalies are flagged whenever the predicted behavior deviates from the actual one. More details about the network will be provided in Section IV-B.

IV. Results and Discussion

A. Explorative Data Analysis

An example of the collected data for different plant conditions is shown in Fig. 2 where three out of the 20 analyzed plants are presented. The top three plots display the relative stem frequency change concerning each plant's initial calibration frequency. The bottom three plots present the corresponding soil water potential data The green background indicates when plant is not showing

TABLE I
BASELINE FREQUENCY VALUES FOR EACH PLANT

Plant ID	Baseline (kHz)	Plant ID	Baseline (kHz)	Plant ID	Baseline (kHz)	Plant ID	Baseline (kHz)
1	21.6	5	25.6	9	16.4	13	13.9
2	21.3	6	23.4	10	13.8	14	12.8
3	23.7	7	13.5	11	13.9	15	13.4
4	20.4	8	22.4	12	13.3	16	13.9
17	15.6	18	14.6	19	13.8	20	17.0

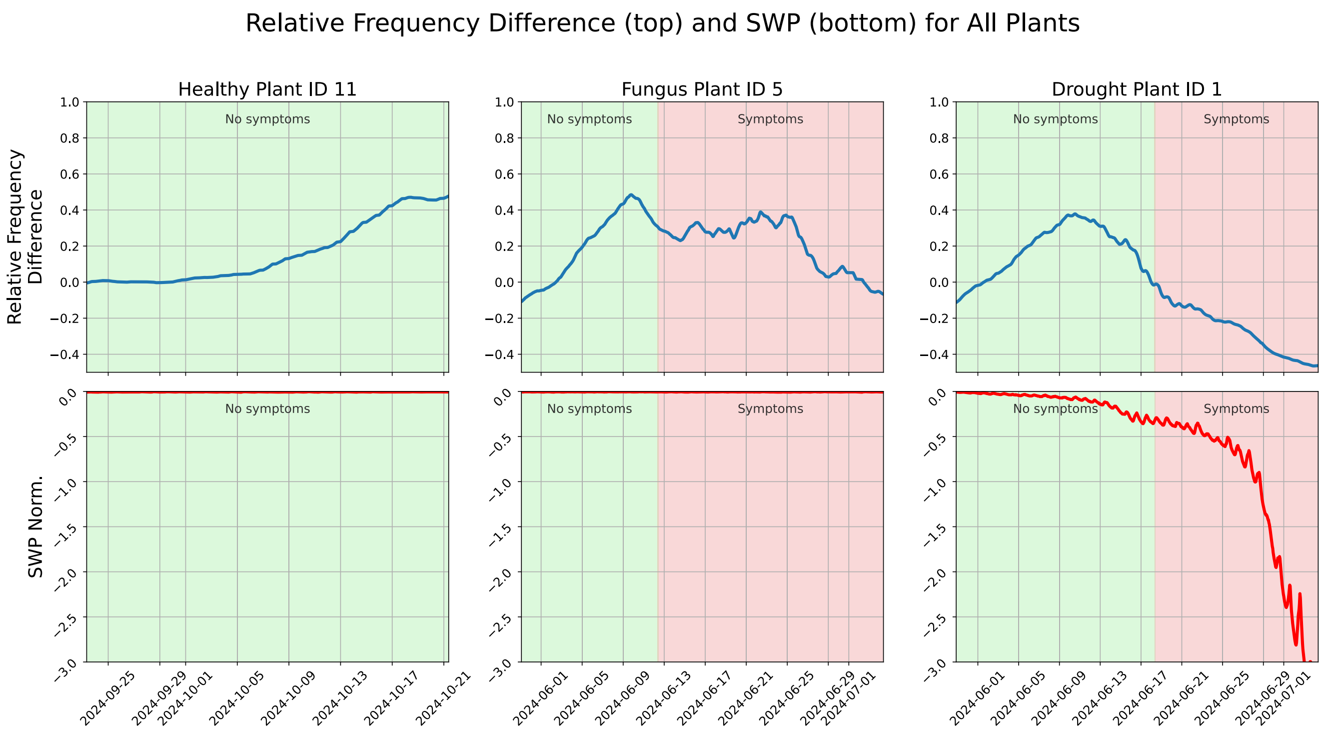

Fig. 2. Example of stem frequency relative change under different conditions. The blue line shows the relative changes in stem frequency, while the red line indicates the corresponding soil water potential. Green background areas denote periods with no visible plant stress symptoms. In contrast, red backgrounds indicate visible stress symptoms (due to drought or Fusarium infection) from left to right: healthy, Fusarium-infected, and drought-stressed plants.

any visible symptom of stress, e.g., drought (abiotic) or Fusarium (biotic).

Upon visual inspection, each tomato plant in Fig. 2 exhibits an initial increase in stem frequency, suggesting a reduction in stem conductivity—likely due to increased stem diameter (analogous to a larger wire having lower resistance and conducting more). In healthy plants (left most plots in Fig. 2), this increase continues until it reaches a plateau, stabilizing except for natural day–night variations [21]. In contrast, plants under drought stress (an example is shown in Fig. 2, right-most plots) initially follow a similar increasing trend but then decline in stem frequency before the first visible symptoms appear (e.g., wilting, yellowing leaves). This decrease suggests a decrease in stem conductivity, as observed in [21]. Similarly, plants infected (an example is shown in Fig. 2, central plots) with the Fusarium pathogen also exhibit an initial increase in stem frequency, followed by a decreasing trend. However, this decline is less pronounced compared to drought-stressed plants. The trend becomes more evident after the first visible symptoms appear, likely due to the progressive occlusion of vascular tissues by the Fusarium pathogen, which worsens over time as the infection spreads.

To better understand the distribution of data for the different plant conditions, the stem frequency features (RFC and stem frequency slope) are represented on the boxplots of Fig. 3. Boxplots represent the distribution of a dataset using five key statistics: minimum (lower end of the left whisker), first quartile (bottom edge of the box), median (line inside the box), third quartile (top edge of the box), and maximum (upper end of the right whisker), with the possibility of highlighting outliers, which are values that fall outside the so-called “whiskers” (the horizontal lines).

Different consideration can be done for the different plots as follows.

- *RFC (leftmost boxplot):* Plants with symptoms exhibit a slightly lower median RFC than healthy plants. However, the interquartile ranges of both groups overlap significantly, suggesting substantial variability in stem frequency

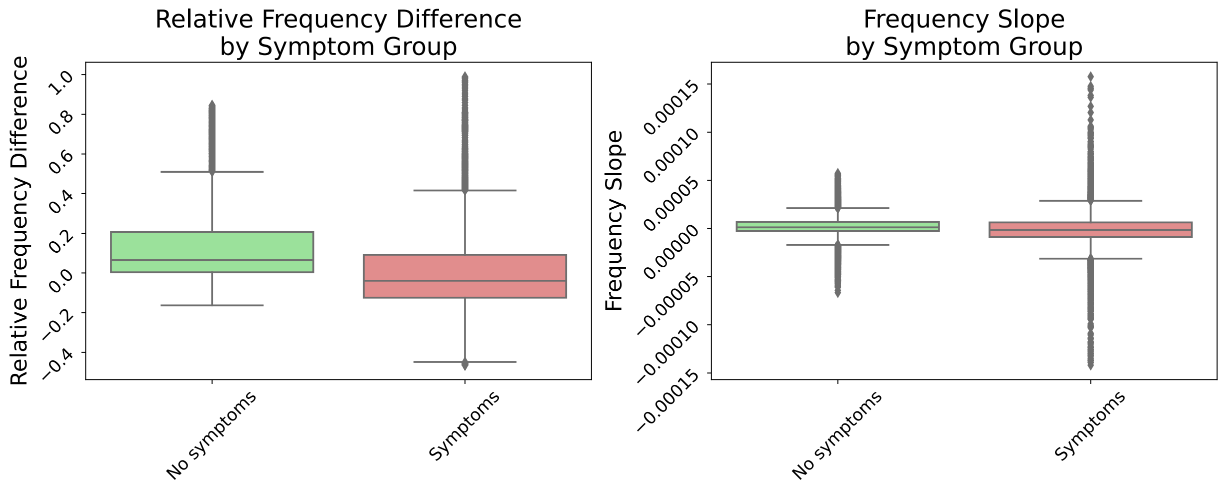

Fig. 3. Boxplot for the different stem frequency features.

and consequently, in stem conductivity, when the plant is subjected to stress.

- *Frequency slope (rightmost plot):* The median frequency slope values are comparable between symptomatic and nonsymptomatic plants. However, symptomatic plants display a broader range and many outliers, indicating sudden fluctuations in stem frequency trends, which may correspond to stress-induced physiological changes.

B. Forecasting and Anomalies

Plants are divided into training and test datasets based on their condition: healthy plants are used for training, while plants experiencing stress are assigned to the test dataset. The input to the LSTM neural network consists of 24 time steps of the selected features

$$X_t = [x_{t-23}, x_{t-22}, .., x_t] \quad (4)$$

and the output of the network is the value of relative stem frequency change for the next day $y_t = x_{t+24}$. Each plant's data are segmented into sequences using a sliding window approach. For the training dataset, which includes eight plants, we obtained a total of 473 sequences per plant, with each sequence consisting of 24 samples (24 hours).

For training and validation, the leave-one-group-out (LOGO) technique [22] is employed. In each iteration, all but one plant from the training dataset is used for training, while the excluded plant is used for validation. This process allows for performance evaluation and hyperparameter tuning. After completing this step, the final model is trained on the entire training and validation samples and then evaluated on the test plants belonging to the stress groups (drought and Fusarium) which are completely unseen from the model.

The trained model is used to predict stem frequency values 24 hours ahead, which are then compared to the actual measured values. An anomaly is flagged if:

1) the prediction error exceeds a predefined threshold (either upward or downward), which accounts for variations in stem frequency that may be fast, but of high value;
2) the actual value remains consistently lower (even for a small amount) than the predicted one from the network for 12 consecutive time steps (equivalent to 12 h).

The second criterion is introduced to capture cases where the model correctly predicts a gradual decline in frequency. In such scenarios, the difference between predicted and actual values may remain below the threshold defined in the first condition, and anomalies would otherwise go undetected. This policy ensures that sustained negative deviations, even if subtle, are still flagged as potential anomalies since, based on prior knowledge, such behavior does not indicate good plant health [21], [17]. Plants exhibiting low or negative RFC (i.e., a decrease compared to their baseline) are likely experiencing growth issues. In contrast, healthy plants typically show an initial increase in stem frequency, followed by stabilization at a specific value. In our case, we set the prolonged threshold value to –0.05 of RFC, indicating a reduction of 5% compared to the initial condition of the plant. A higher threshold (15%) is set for anomalies in which the actual frequency value is higher than the predicted ones, since higher stem frequencies (hence higher stem conductivity) are usually associated with better plant hydration and activity [17], [21].

The employed network was a single-layer LSTM neural network with one hidden layer with 20 neurons. The input LSTM layer processes the input sequence, capturing temporal patterns across time steps. It outputs a sequence of hidden states, one for each time step in the input. Then, the last hidden state of the LSTM is passed to a fully connected layer to output the prediction. The training was performed during 60 epochs, with a batch size 32 and a learning rate of 0.01. The employed loss function is Pytorch's mean squared error loss. The results for training and validation losses for one of the plants during the LOGO procedure are reported in Fig. 4.

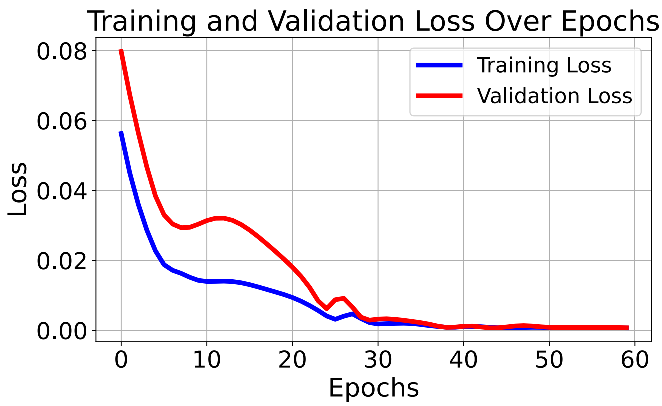

Fig. 4. Training and validation loss over epochs for LSTM network.

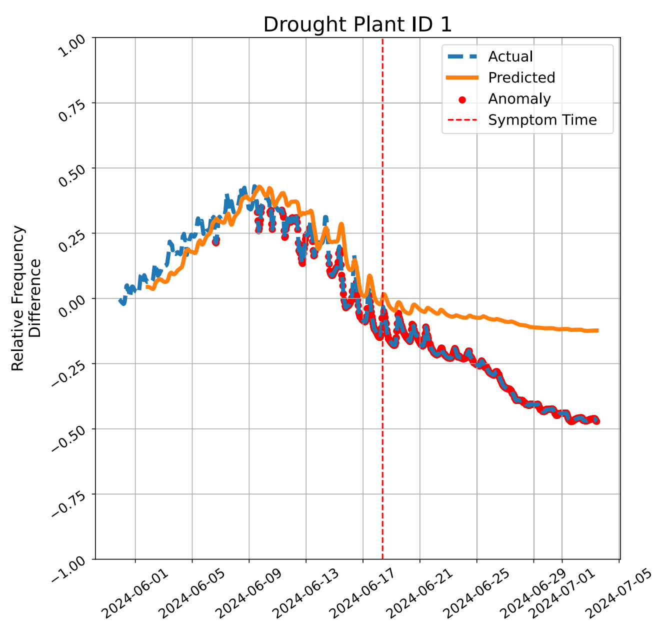

Fig. 5. Results of LSTM predictions for a plant under drought stress, highlighting detected anomalies. The blue dashed line represents the actual measured values, the orange line shows the network's predicted values, and the red markers indicate detected anomalies.

After the training, process the LSTM model was used on symptom plants according to the anomaly detection policy selected.

Fig. 5–7 suggest that the network is able to capture patterns that approximate the plant's behavior. However, at a certain point (Fig. 5), the actual values deviate markedly from the training examples, leading to an increased prediction error and the detection of an anomaly. Notably, some anomalies are detected even before visible symptoms appear in the plant (Fig. 5). This highlights the potential of using stem frequency, combined with anomaly detection techniques, for early detection of drought stress in plants.

Fig. 6 presents the results obtained for a Fusarium-infected plant. In this case, the anomaly detection was less effective compared to drought stress. While the algorithm successfully identifies a sudden shift in the RFC before the first visible symptoms appear, it fails to consistently flag most of the samples collected when the plant is under stress. As a result, many stress-related data points are not correctly classified as anomalies.

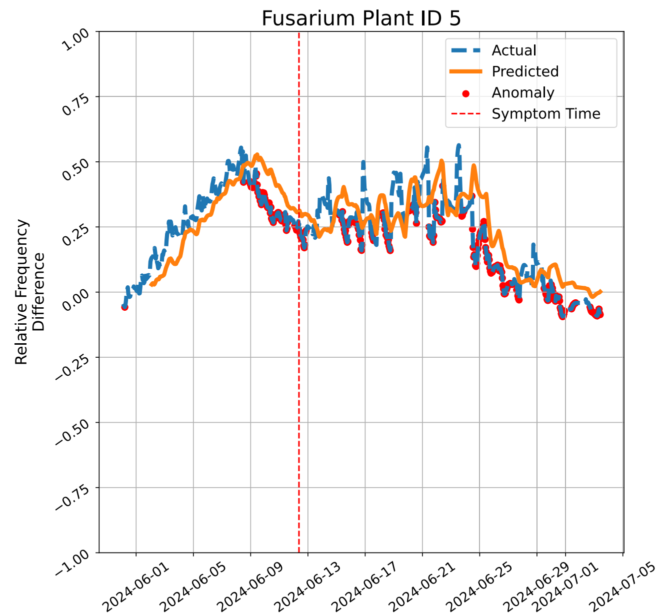

Fig. 6. Results of LSTM predictions for a plant under biotic Fusarium stress, highlighting detected anomalies. The blue dashed line represents the actual measured values, the orange line shows the network's predicted values, and the red markers indicate detected anomalies.

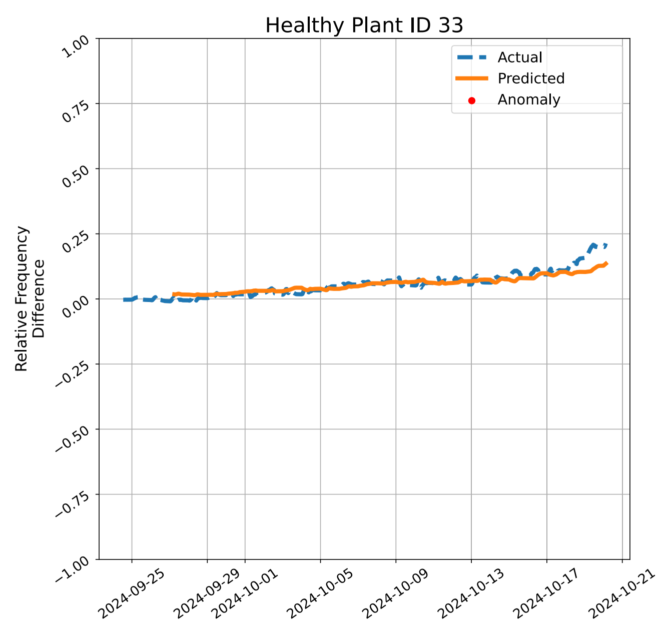

Fig. 7. LSTM prediction results for a healthy plant, illustrating detected anomalies. The blue dashed line represents the actual measured values, the orange line corresponds to the network's predicted values, and the red markers indicate detected anomalies.

The training was repeated while leaving out a plant in healthy condition to validate the anomaly detection approach further. In doing so, we ensured the procedure did not incorrectly flag false anomalies. The results of this experiment are presented in

TABLE II
ANOMALY DETECTION METRICS FOR EACH PLANT, GROUPED BY CONDITION (DROUGHT OR FUSARIUM)

Plant ID	Condition	Precision	Recall	F1-score
1	Drought	1.00	0.99	1.00
2	Drought	1.00	0.93	0.96
3	Drought	1.00	0.74	0.85
4	Fusarium	1.00	0.61	0.76
5	Fusarium	1.00	0.53	0.69
6	Fusarium	1.00	0.28	0.44
8	Fusarium	1.00	0.47	0.64
14	Drought	1.00	1.00	1.00
15	Drought	1.00	1.00	1.00
16	Drought	1.00	1.00	1.00
18	Drought	1.00	1.00	1.00
19	Drought	1.00	1.00	1.00

TABLE III
TIME DIFFERENCE BETWEEN THE FIRST DETECTED ANOMALIES AND THE FIRST VISIBLE SYMPTOMS FOR EACH PLANT

Plant	Condition	Days Before Symptoms
1	Drought	7
2	Drought	4
3	Drought	2
4	Fusarium	1
5	Fusarium	1–2
6	Fusarium	2
8	Fusarium	4
14	Drought	6
15	Drought	4
16	Drought	1
18	Drought	1
19	Drought	2

A positive value indicates that anomalies were detected before symptoms became visible.

Fig. 7. The trained model can capture the stable upward trend in relative frequency over time, which is expected due to plant growth.

Since the testing dataset includes multiple plants, presenting an individual plot for each one is impractical. Therefore, we evaluated precision and recall for anomaly detection (i.e., stress symptoms), with the results summarized in Table II. This evaluation considers only samples recorded after the first appearance of symptoms. Symptoms are always present in this subset, but the model does not always detect them. As a result, precision remains 1.00 for all plants, as there are no false positives, i.e., every sample in this period represents a true anomaly.

However, recall varies significantly across different plants, meaning that the model's ability to identify symptomatic cases correctly is inconsistent in all stress conditions. Most plants with drought symptoms (1, 2, 3, 14, 15, 16, 18, 19 exhibit high recall values, indicating that the model effectively captures drought stress symptoms. Conversely, the algorithm struggles with Fusarium-affected plants (4, 5, 6, 8), leading to lower recall values. Since every sample after symptom onset is considered an anomaly, precision is always 1, making recall the most relevant metric for evaluating the model's performance. In this context, recall directly represents the proportion of actual anomalies that were correctly identified. The overall recall for drought-stressed plants is 0.89, while for Fusarium-affected plants, it is 0.49, highlighting the problems of detecting biotic stress symptoms compared to abiotic stress.

However, even if symptoms are not always correctly identified in biotic stress conditions, the early detection of flagged anomalies can serve as an early warning for the future onset of stress symptoms. The algorithm detects abnormal behavior before symptoms become visually apparent, providing a valuable tool for proactive monitoring.

For this reason, we also report in Table III the time point at which a plant accumulates a certain number of detected anomalies specifically, 12 h with flagged symptoms.

In most cases, this time threshold is reached before the first noticeable visible symptoms appear, suggesting that the model can provide early indications of plant stress even when recall is not perfect. Particularly interesting are the results for plants 1, 2, and 14 (drought), where anomalies were noticed seven, four, and six days before the first visible symptoms on the plant, respectively. For Fusarium, on average the anomalies were identified one day before visible symptoms but many anomalies are not identified at all as indicated by the low recall.

V. CONCLUSION AND FUTURE WORKS

In this study, we evaluated the ability of an anomaly detection LSTM model to identify plant stress conditions based on recorded symptoms. Our results indicate that the model performs well in detecting drought-induced stress, achieving high recall and F1-scores across most plants. The overall F1-score for drought-affected plants was 0.93, making the model effective in capturing drought-related symptoms. Conversely, the model struggled with identifying Fusarium-induced stress, with a lower overall F1-score of 0.65. This suggests that biotic stress symptoms may present more subtle or variable patterns that are harder to detect using the current approach. Despite the lower recall for Fusarium, the early detection of anomalous behavior could serve as a warning signal before visible symptoms appear. By analyzing the time at which a sufficient number of anomalies were detected, we found that for most plants, early warnings were triggered several days before the first visible symptoms. This highlights the potential of anomaly detection methods as a proactive monitoring tool for plant health, allowing early interventions before stress conditions become severe. Future work should explore refinements in the model, such as incorporating additional physiological and environmental parameters, to improve sensitivity to biotic stress factors. In addition, testing the approach on a larger dataset with different plant species and stress conditions would help assess its generalizability and robustness.

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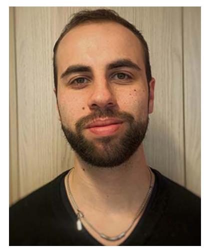

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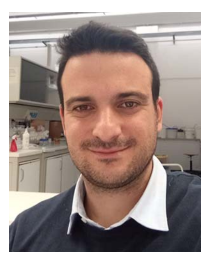

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